M12 Dopamine

Scope of Practice	AP
Classification	• Catecholamine, Positive Inotrope, Vasopressor
Usual Presentation	• 200mg/5ml ampule • **Note: Ampule presentation in mg, dose calculation in mcg**.

Mechanism of Action	• Dopamine Stimulates Dopaminergic (D1 & D2), Adrenergic (α1 & ß1) receptors, causing:
	  • **At Low Doses** (1–5 mcg/kg/min IV/IO):Dopamine Receptors (D1 and D2): increased blood flow to the kidneys and improving urine output.
	  • **At Medium Doses** (5–10 mcg/kg/min IV/IO): Beta-1 Adrenergic Receptors: increased heart rate and contractility (inotropic effect), improving cardiac output.
	  • **At High Doses** (10–20 mcg/kg/min IV/IO): Alpha-1 Adrenergic Receptors: causing vasoconstriction, increasing blood pressure.

Indications	• Hypotension/Shock associated with:
	  • Cardiogenic shock / Congestive Heart Failure
	  • Bradycardia
	  • Post cardiac arrest
	  • Sepsis

Contraindications	• Known allergy • Pheochromocytoma
Precautions	• Arrhythmias are associated with high doses required for vasoconstriction
Side Effects	• Arrhythmias (particularly with high doses – including VF). • Chest pain • Palpitations • Widening of the QRS complex
Adult Dose & Administration	• 2-20**mcg**/kg/min IV/IO • Use syringe driver wherever possible
Pediatric Dose & Administration	• 2-20**mcg**/kg/min IV/IO • Use syringe driver wherever possible.
Use in Pregnancy and Lactation	• Pregnancy: May cause fetal harm. However, dopamine should be administered where clinically indicated. • Lactation: May be excreted in breast milk. However, dopamine should be administered where clinically indicated.
Additional Information	• Due to difficulties in administration without a syringe driver, carefully evaluate the requirement for dopamine rather than using an alternative (e.g. adrenaline) which may be easier to administer accurately.